

Beyond the Syntax: Mastering Agentic Coding

The Future of Software Engineering

Awwal Badru

PhD Candidate & MS, CS (United States) | M.Sc CS (Australia)

| B.Ed CS/Maths (Nigeria)

Research Areas:

Empirical Software Engineering
Software Security
Software Engineering Education
Computer Science Education



vs



vs



Workshop Outline

DAY 1: Foundations & Demo

- The Evolution of AI & Coding
- Coding Agents & Agentic Loop
- Demo: Context, Action & Verification
- IDE Setup and Resources
- Interactive Q & A

Workshop Outline

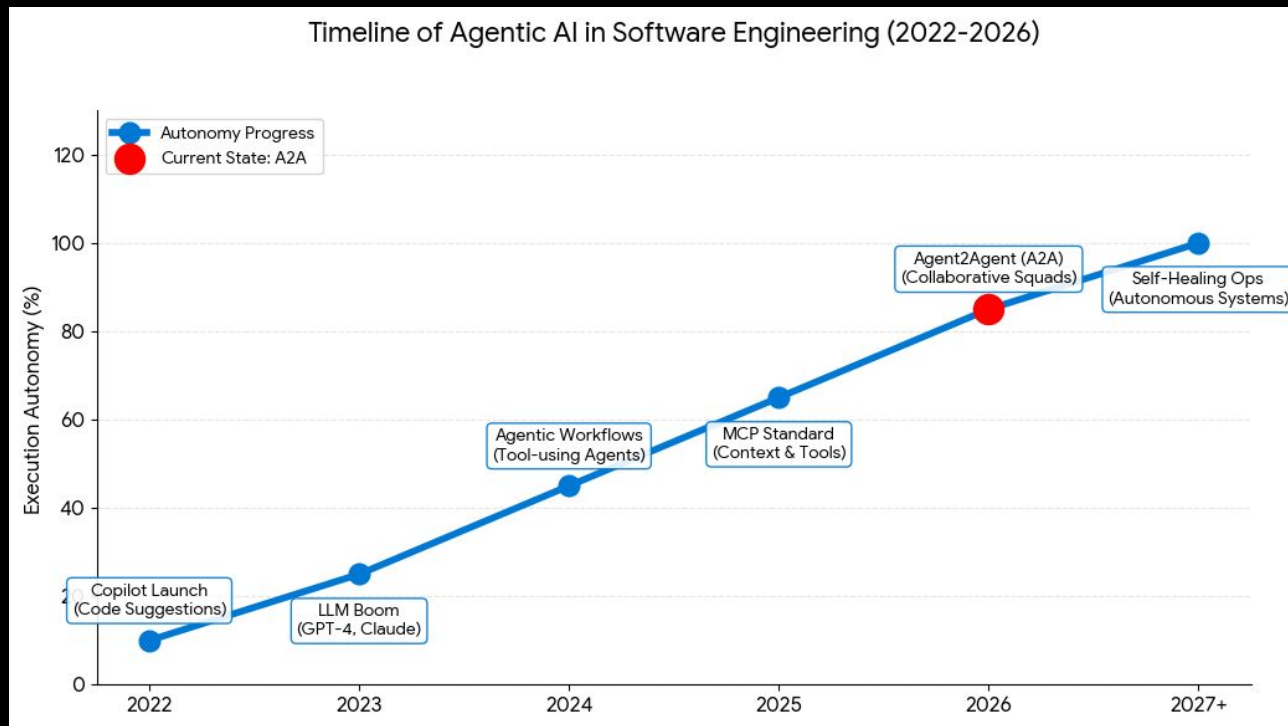
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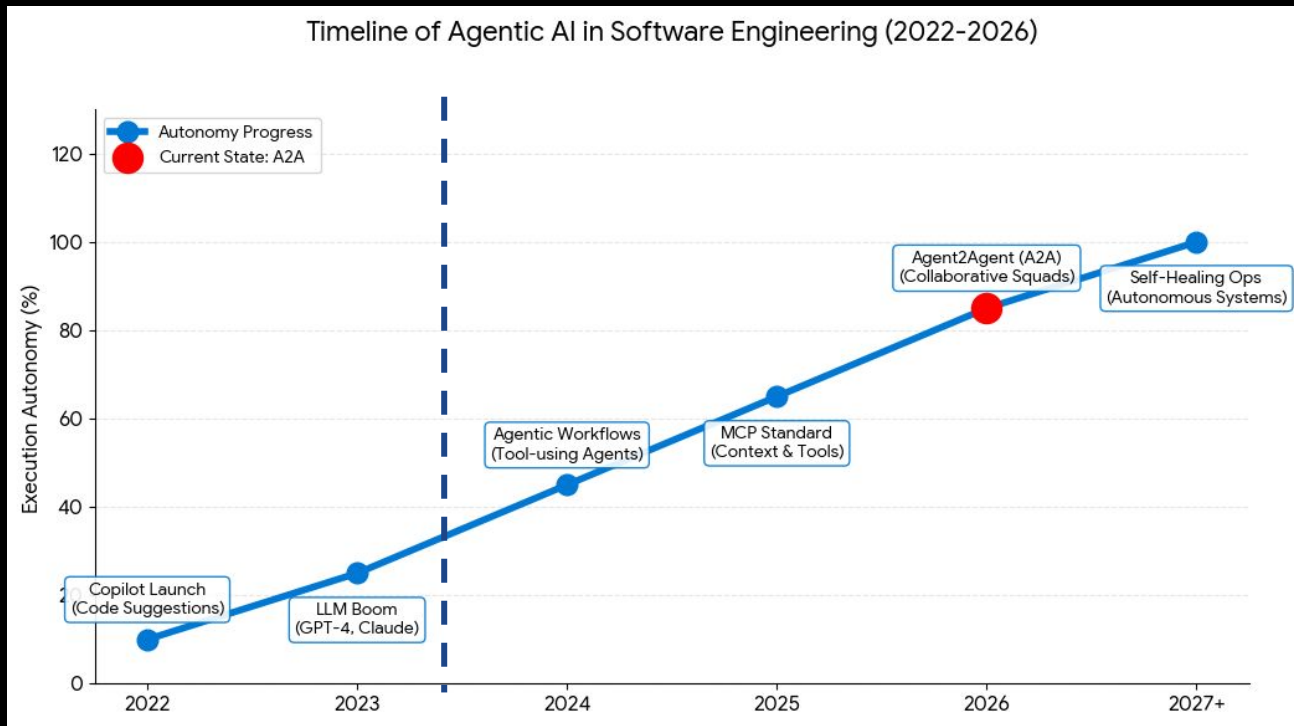
DAY 2: Practical Application

- Software Development Life Cycle
- Hands-on Building with Agents
- Git Versioning & Collaboration

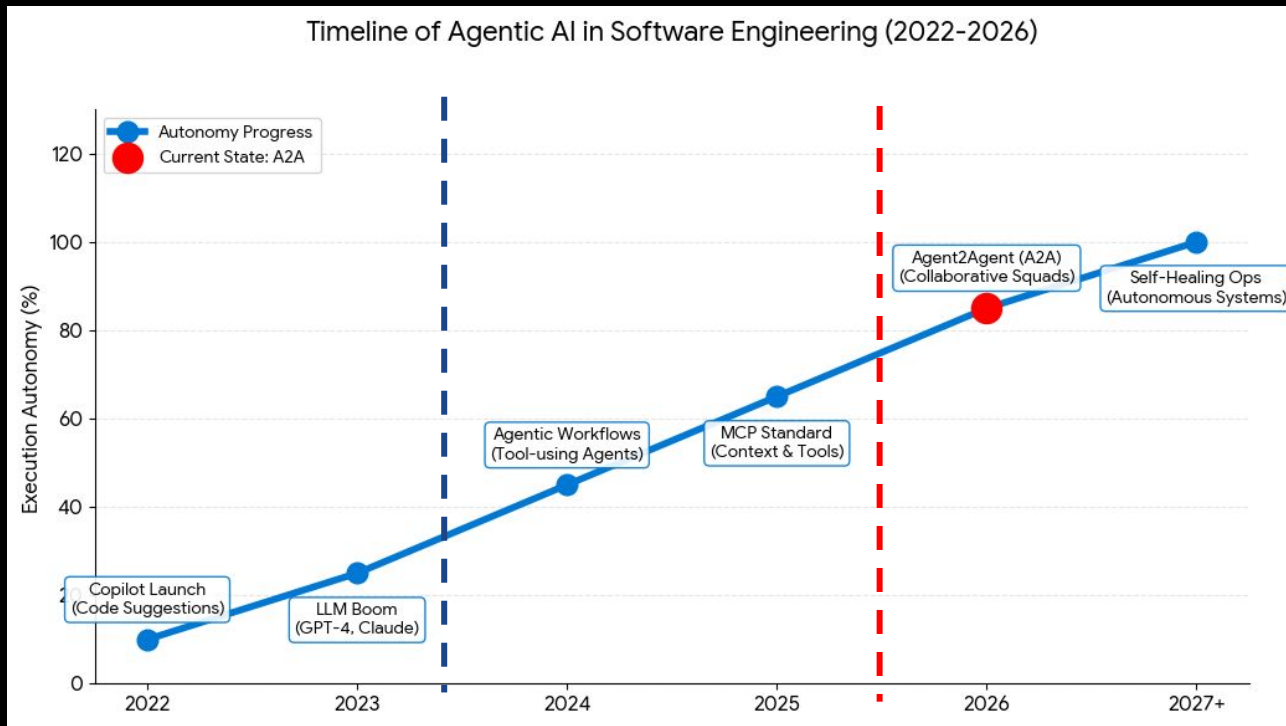
The Evolution of AI & Coding



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The Evolution of AI & Coding

1

Code Completion

Intelligent snippets and autocomplete based on context. Tools like traditional IDE intellisense.



The screenshot shows a code editor with a file named 'hello_world.py'. The code is as follows:

```
1 def hello_world():  
2     print("Hello, World!")  
3  
4 if   
5     hello_world()
```

A red box highlights the 'hello_world()' suggestion in the autocomplete dropdown menu. The dropdown menu also shows 'Accept Word' and 'Accept' buttons.

The Evolution of AI & Coding

2

Code Generation

Generative AI creating entire functions and boilerplate from natural language prompts. The era of Copilots.

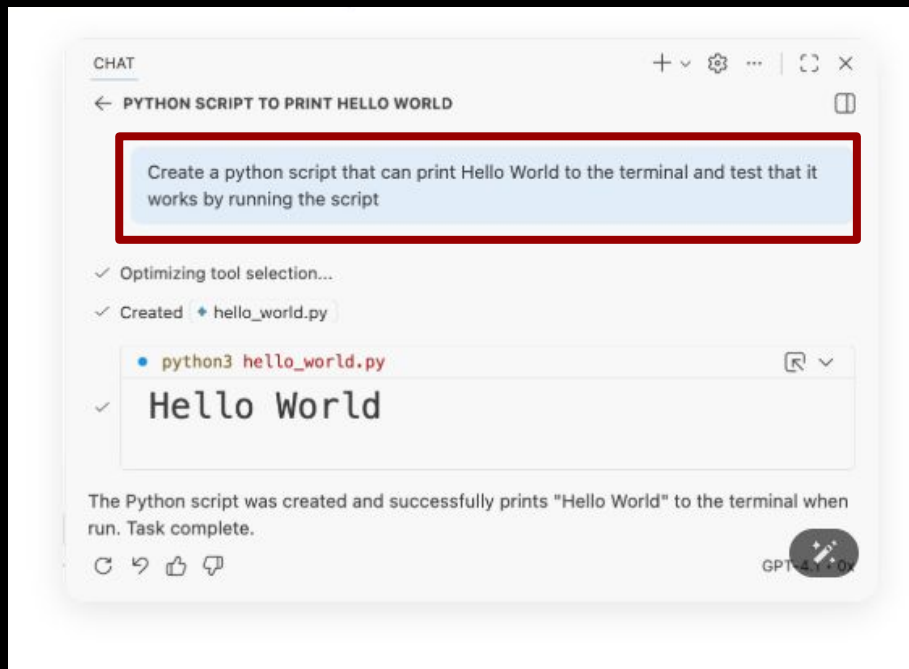


The Evolution of AI & Coding

3

Agentic Coding

Autonomous agents that reason, plan, and execute across files, using tools and verifying results.



Coding Agents

IDE

- Copilot in VS Code
- Cursor
- Windsurf
- Antigravity

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- Copilot in GitHub.com
- Codex
- Jules

Coding Agents



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Custom

- Copilot SDK
- PyDantic AI
- LangChain

Coding Agents & Agentic Loop

What is an AI Agent?

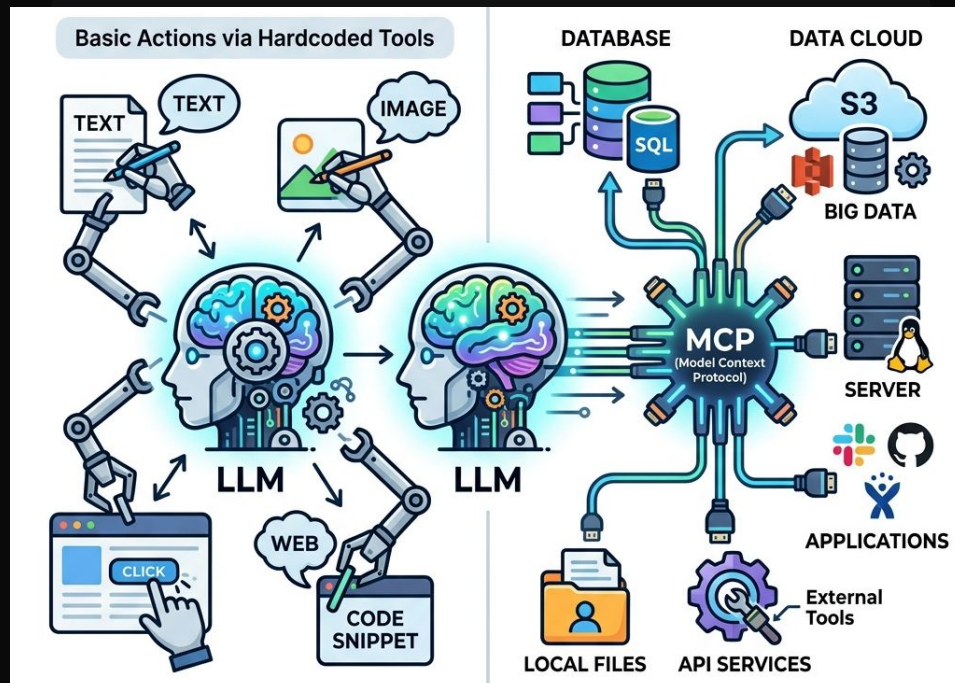
An AI Agent is an LLM that calls tools in a loop to achieve a goal.

Source: https://simonwillison.net/2025/Sep/18/agents/?utm_source=chatgpt.com

Coding Agents & Agentic Loop

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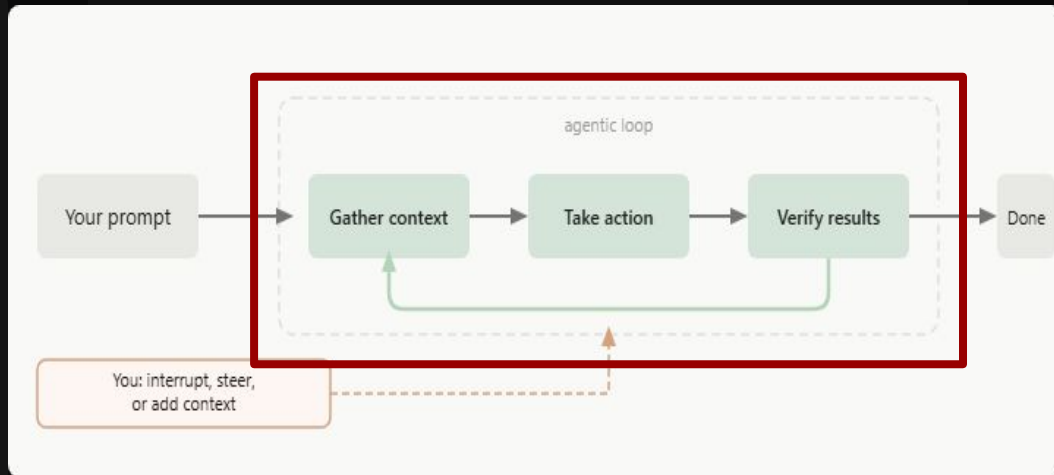
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Coding Agents & Agentic Loop

What is an AI Agent?

An AI Agent is an **LLM** that **calls tools** in a **loop** to achieve a goal.



Source: <https://code.claude.com/docs/en/how-claude-code-works>

Quick Pricing Overview (April 2026)

Service	Individual Plan	High-Usage/Max	Team/Business
GitHub Copilot	\$10 /mo	\$39 /mo (Pro+)	\$19 /user/mo
Claude	\$20 /mo	\$100 - \$200 /mo	\$25 /user/mo
Antigravity	\$29 /mo	\$99 - \$250 /mo	Custom
OpenAI Codex	\$20 /mo (via ChatGPT Plus)	\$200 /mo (Pro)	\$25 /user/mo

Demonstration

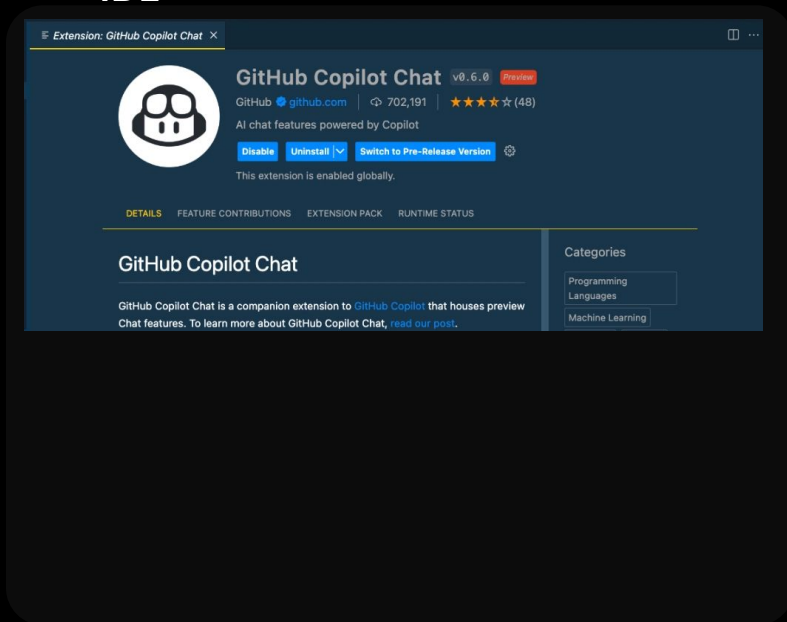
 IDE

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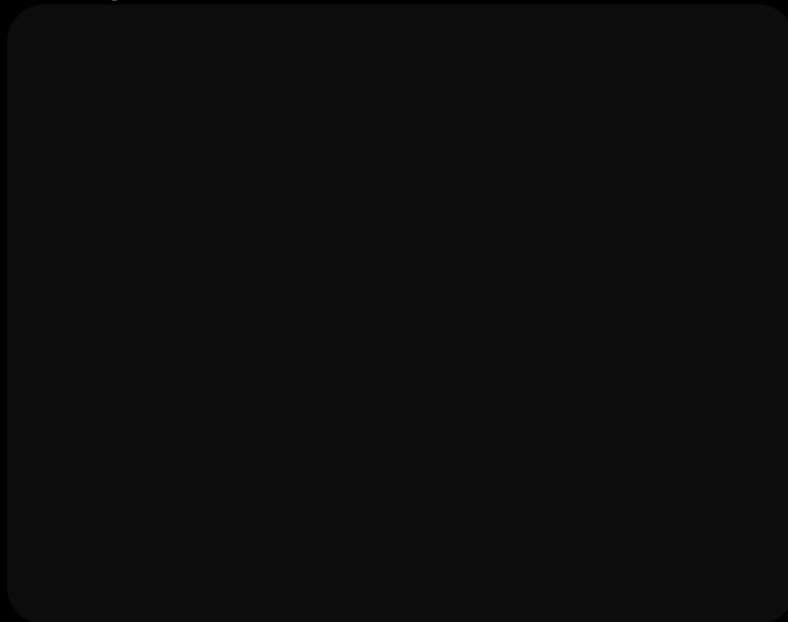
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IDE



CLI



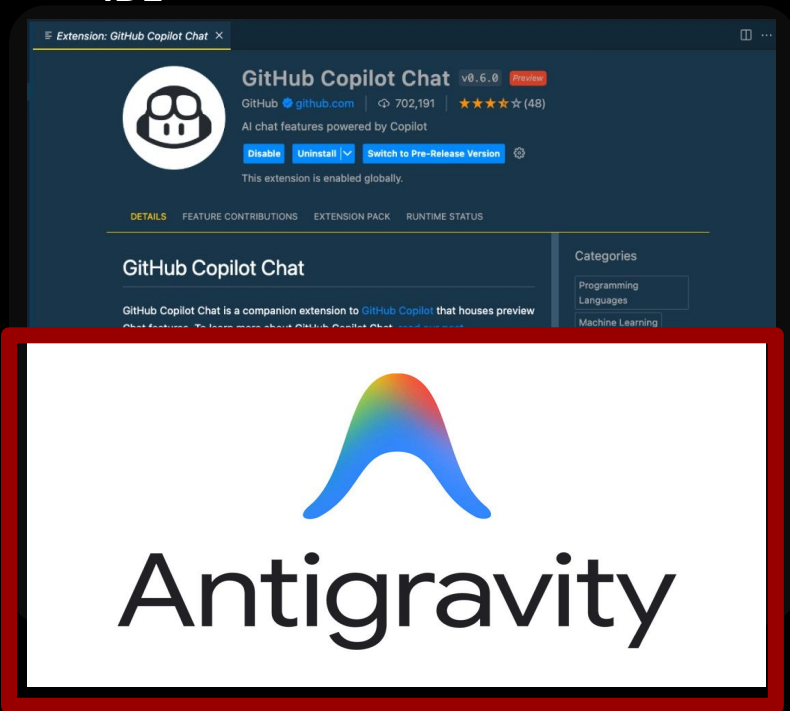
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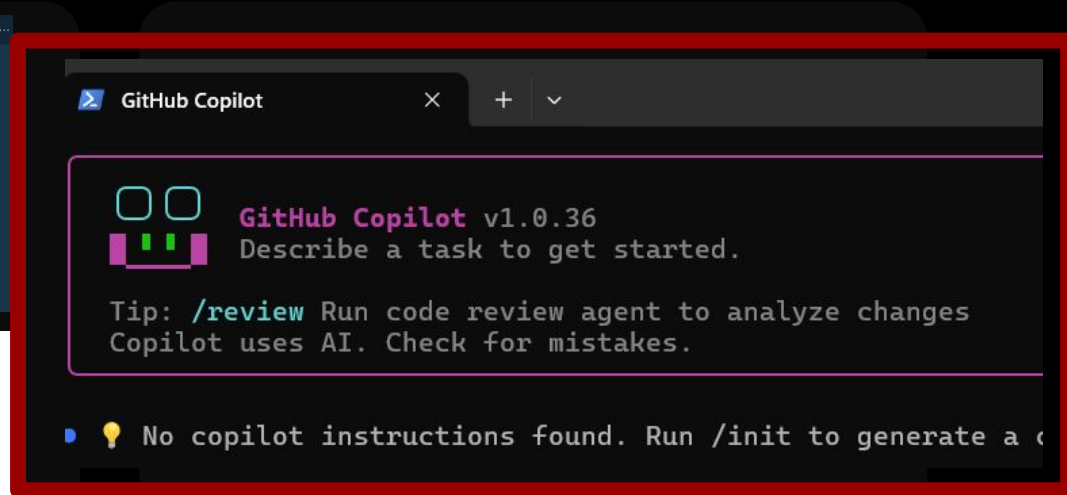
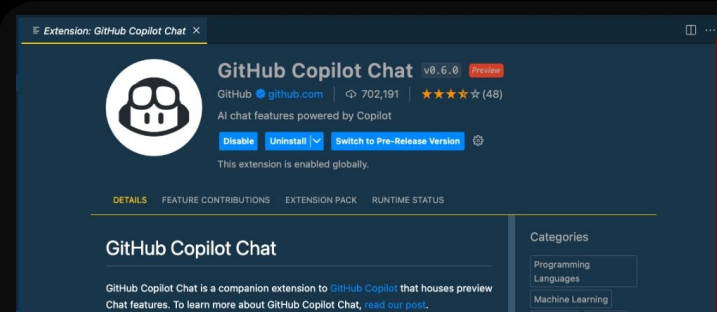
IDE



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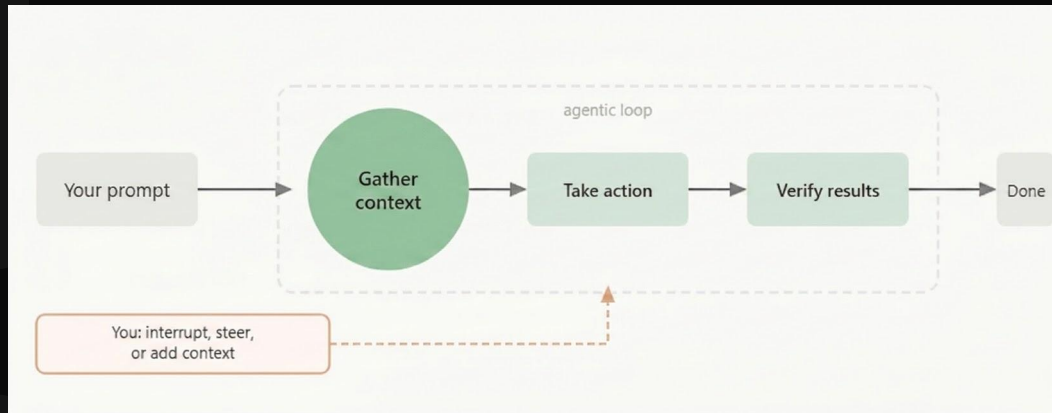
Demonstration



Equipping Agents with Context

Including Files and Assets

- Directly provide files, images, and GitHub items (Issues, PRs)
- Offer exemplar files containing best-practice code



Directive Documentation

- `copilot-instructions.md` — GitHub Copilot
- `cursor rules` — Specific to Cursor IDE
- `claude.md` — Tailored for Claude Code
- `agents.md` — The standard OpenAI format

Providing Tools to an Agent

MCP — Model Context Protocol

Open protocol for giving agents access to tools

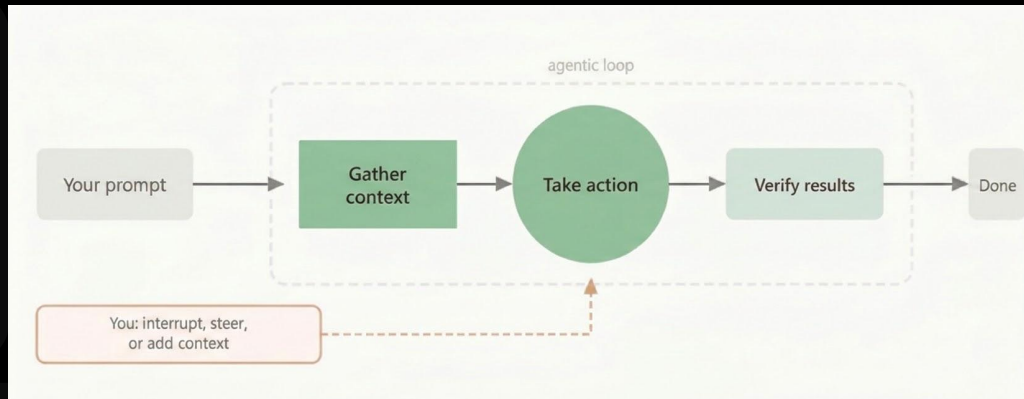
In IDE: search @mcp in extensions for vetted servers

Pro Tip: You can also create your own MCP servers

Agent Skills

A reusable workflow defined in a `SKILL.MD` file

Skills can include tools, MCP servers, scripts, and more!



Resources

IDE (VS Code)

- VS Code Installation
<https://code.visualstudio.com/download>
- Copilot Installation from VS Code extension
<https://www.youtube.com/watch?v=ws6t0NBle9g>
- Launching Copilot for free (please you need to have a github account)
https://www.youtube.com/watch?v=X_Aet9ndh_Y

IDE (Antigravity)

- Google Agent IDE Installation
<https://antigravity.google/download>

You need to have gmail account to sign in and start using the agent for free

Resources

CLI (Copilot)

- Installation npm (requires Node.js 22+)

<https://www.youtube.com/watch?v=BDxRhhs36ns>

- Installation on windows via (WinGet)

https://www.youtube.com/watch?v=3mCaZ_tYdI0

CLI (Copilot)

Install commands:

Windows (WinGet): winget install
GitHub.Copilot

npm (requires Node.js 22+): npm install
-g @github/copilot

macOS / Linux: brew install copilot-cli

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The 7 Phases of SDLC

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6. Deployment

Release to production environment.

7. Maintenance

Fix issues, update, and monitor.

Hands-on Learning

- For UX/UI

<https://stitch.withgoogle.com/>

Hands-on Learning

What app are you building?

1. Planning

Define scope, costs, and identify risks.

2. Requirements

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3. Design

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